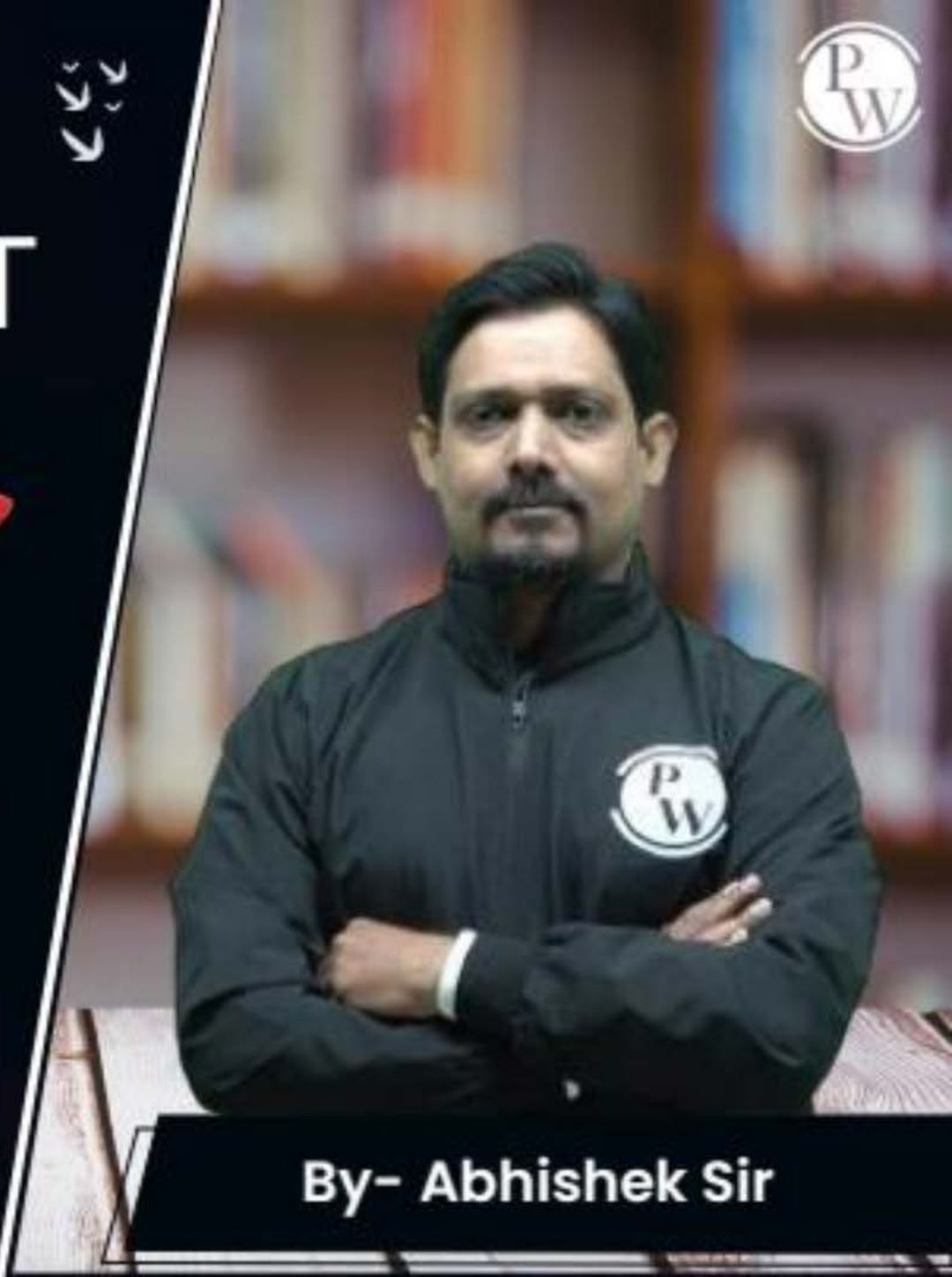
Computer Science & IT



C programming

Data Types & Operator

Lecture No. 01



By- Abhishek Sir

Topics to be Covered



- Topic — Structure of C program
- Topic — type of statement
- Topic — Tokens in C-language
- Topic — Constant
- Topic

Slide



Structure of C program

```
# include<stdio.h>
int main() {
    int a=10, b;
    int c,
    b = 20;

    c = a+b;
    printf("%d", c);

    return 0;
}
```

header file

stdio — standard input output file

include — reusable code

↑

macro



Structure of C program

```
# include<stdio.h>
int main() {
    int a=10, b;
    int c,
    b=20;

    c=a+b;
    printf("%d", c);
    return 0;
}
```

return type

Name of funchm

int main() ← main function

When any C program Run

then program execution

Start form main

return 0 means Successful execution of program

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Structure of C program

```
# include<stdio.h>
int main() {
    int a=10, b;
    int c,
    b = 20;

    c = a+b;
    printf("%d", c);
    return 0;
}
```

{ Body of function

Local variable

Scope and life of variable (a, b, c)

} Scope defins which of part of program variable can be accessed, modified

Slide



Structure of C program

```
# include<stdio.h>
int main() {
    int a=10, b;
    int c,
    b = 20;

    c = a+b;
    printf("%d", c);

    return 0;
}
```

within bracket all variable will be Local

Slide



Structure of C program

```
# include<stdio.h>
int main() {
    int a=10, b;
    int c;
    b = 20;
    c = a+b;
    printf("%d", c);
    return 0;
}
```

statement of C Language

Semicolon ;

Slide



Structure of C program

```
# include<stdio.h>
int main() {
    int a=10, b;
    int c,
    b=20;
    c=a+b;
    printf("%d", c);
    return 0;
}
```

Declarative Statement

int c; ← declarative statement

c is integer variable

int a = 10;

↑

Declaration & initialization

↑

Initial value

Slide



Structure of C program

```
# include<stdio.h>
int main() {
    int a=10, b;
    int c,
    b=20;
    c=a+b;
    printf("%d", c);
    return 0;
}
```

Arthmetic & logical statement

b=20;
↳ equal to

c=a+b
↳ addition

arthmetic operators

Slide



Structure of C program

```
# include<stdio.h>
int main() {
    int a=10, b;
    int c,
    b=20;

    c=a+b;
    printf("%d", c);
    return 0;
}
```

→ Input output statement

printf("%d", c);

Name of function ↑

parameter ↑

format specifier

%d - integer

integer < +ve, -ve

Slide



Structure of C program

```
# include<stdio.h>
int main() {
    int a=10, b;
    int c,
    b=20;

    c=a+b;
    printf("%d", c);
    return 0;
}
```

return - terminates the program

return 0 Signifies

Successful execution of program



Structure of C program

Type of statement

1. Declarative
2. Arithmetic & logical
3. Input & output
4. Control statement

Slide



Structure of C program

Statement of c program will further divided to small units called token.

```
int a = 10;
```

int → integer keyword : Reserve words that can be used

Slide



Structure of C program

Statement of c program will further divided to small parts called token

```
int a = 10;
```

a ⟶ Name of variable :. Identifier



Structure of C program

Statement of c program will further divided to small parts called token

```
int a = 10;
```

→ operator token

Arithmetic, Relational

Logical operator, Bitwise

Slide



Structure of C program

Statement of c program will further divided to small parts called token

int a = 10; → Numeric constant.

Slide



Structure of C program

Statement of c program will further divided to small parts called token

```
int a = 10;
```

Special character

Slide



Structure of C program

```
#include<stdio.h>
int main() {

    int a=10, b;
    int c;
    c=a+b;
    printf("%d", c);
    return 0;
}
```

Slide



Topic : Token in C language



Slide

Name of a variable : Identifier must start with

underscore (_) or alphabet followed by underscore (_)

alphabet (a...z, A...z) or digit (0....9)

① 6_day ③ _init ⑤ 64_id X ⑦ _@dept

② dep-36 X④ while ⑥ 6ac-d X ↑ @ Not allowed

↑ Reserve keyword ↑ Numeric

⑧ Int ← Name of identifier

↑ yes

C is case sensitive language



Topic : Reserve keyswords

auto	double	int	struct
break	else	long	switch
case	enum	register	typedef
char	extern	return	union
const	float	short	unsigned
continue	for	signed	void
default	goto	sizeof	volatile
do	if	static	while



Topic : Constant

1. Decimal (0...9) $123 = 1\times10^2+2\times10^1+3\times10^0$

2. Binary (0,1) : $11101 = 1\times2^4+1\times2^3+1\times2^2+0\times2^1+1\times$
$= 16+8+4+0+1 = 29$

3. octal (0..7) : $56 = 5\times8^1+6\times8^0 = 40+6 = 46$

4. Hexadecimal : $A8 = A\times16^1+8\times16^0$
$10\times16+8 = 168$

(0..9, a, b, c, d, e, f)

Slide



Topic : Constant

```
int a = 10;
```
← Decimal

```
int a = 0b10;
```
zero followed by b

↑ Binay constant

```
int a = 010;
```
→ zero first digit is zero

octal Constant

```
int a = 0x11;
```
0 followed by x is hexadecimal

Slide



Topic : Constant

```
#include <stdio.h>
int main(){
    int a = 15;
   int b = 0b101;
   printf("%d", a+b);
    return 0;
}
```

A. 94
B. 272
C. 20
D. 80

[C]

$101_{210} = 1\times2^2 + 0\times2^1 + 1\times2^0$

$4 + 0 + 1 = 5$

Slide



Topic : Constant

```
#include <stdio.h>
int main(){
    int a = 15;
   int b = 0101;
   printf("%d", a+b);
    return 0;
}
```

A. 94
B. 272
C. 20
D. 80

0 1 0 1
2 1 0

65+15=80

$1\times8^2 + 0\times8^1 + 1\times8^0 = 64 + 1 = 65$

Slide



Topic : Constant

```
#include <stdio.h>
int main(){
    int a = 15;
    int b = 0x101;
    printf("%d", a+b);
    return 0;
}
```

Small x

15 + 257

272

A. 94
B. ~~272~~
C. 20
D. 80

0x101 (Hexadecimal)

2 1 0

$1 \times 16^2 + 0 \times 16^1 + 1 \times 16^0$

256 + 0 + 1 = 257

Slide



Topic : Constant

```
#include <stdio.h>
int main(){
    int a;
    a = 29+0b1001+056+0xa;
     printf("%d", a);
    return 0;
}
```

$5\times8^1+6\times8^0$

46

octal → Hexadeimal

10×16^0

1001

3210

8+1=9

29+9+46+10

94

2

A. 94 ✓

B. 272

C. 20

D. 80

[A]

Slide



Topic : Constant

```
#include <stdio.h>
int main(){
    int a;
    a = 60+0b1111+014+0x10;
     printf("%d", a);
    return 0;
}
```

A. 94
B. 272
C. 103
D. 80

Slide



2 mins Summary

Topic — Structure of C program

Topic — type of statement

Topic — Tokens

Topic — Identifier, keywords, constant.

Topic

Slide

THANK - YOU